

LAB EXERCISE - 7

NutriGenomeDB: Nutritional phenotype database

Aim:

The aim of this experiment is to query the NutriGenomeDB database to find and explore how specific genes of interest are differentially expressed in response to various nutrients or bioactive food compounds across a curated collection of nutrigenomics experiments.

Methodology:

1. Go to NutriGenomeDB website, Next click gene expression browser.

The screenshot displays the NutriGenomeDB website. At the top, there is a header with the site name and navigation links: "Gene expression browser", "Phenotype-centered analysis", and "Help". Below the header, a paragraph defines nutrigenomics and mentions the database's use of publicly available gene expression data. A red text update states: "15/11/2021: Upgrade! 56 new nutrigenomics experiments added to the database (287 in total)".

Highlights of NutriGenomeDB platform:

- **Gene expression browser module**
This module is based on a set of manually curated differentially expressed genes after treatment with nutrients or bioactive food compounds obtained from 231 nutrigenomics experiments. All the experiments are linked to their original source in GEO database. For inclusion in NutriGenomeDB, experiments were selected following specific criteria: cell based assays (human cells) from microarray platforms, experimental design with at least 2 replicates per group. This module can be queried by multiple Gene Symbols to look up for differential gene expression values among nutrigenomic experiments, and the results are linked to the original data source. An interactive Lineplot visualization can be obtained, allowing visual exploration of specific genes among NutriGenomeDB experiments.
- **Phenotype-centered analysis module**
This module allows the direct comparison of an external gene expression profile characterizing a phenotype with the experiments hosted in NutriGenomeDB database. It is based on the well documented GSEA algorithm, and its main purpose is to find connections between the introduced

On the right side of the highlights, a vertical flowchart illustrates the data processing pipeline:

- Identification of Nutrigenomics studies
- Experiment annotation
- Gene differential expression analysis
- NA removal, duplicate merging and adjusted pvalue sorting
- Addition to the gene expression database

The flowchart includes visual elements: a "GEO" box, a "target vs control" bar chart, a line plot, and a table of gene expression data.

2. Locate the "Search for Gene Expression data in the database" section on the homepage or Gene Expression Browser. In the provided text box, type the list of Gene Symbols you wish to investigate. Click the Submit button.

NutriGenomeDB

[Home](#) [Gene expression browser](#) [Phenotype-centered analysis](#) [Help](#) A nutrigenomics exploratory and analytical platform

Search for Gene Expression data in the database

e.g. NPC1L1;APOA1

Generate expression heatmap

e.g. NPC1L1;APOA1

Results:

NutriGenomeDB

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Search for Gene Expression data in the database

e.g. NPC1L1;APOA1

Excel

PDF

Search:

Warning: the genes included in the database represent the top 10% differentially expressed features in each experiment. However not all of them are statistically significant under the criteria adj.p-value < 0.05. From a total of 568.463 stored rows in NutriGenomeDB, 156.374 are statistically significant at an adjusted P-value equal or lower than 0.05, corresponding to 27.51% of statistically significant genes among the hosted nutrigenomics gene sets.

Symbol	Log2 FC	Average Expr	p-value	Adjusted p-value	Experiment (GEO id, Cell type, Treatment, Concentration, Duration)
NETO2	100.0	100.0	0.000	0.028	GSE23984 CD3+T VitaminD analog 10-8M 8days
NETO2	100.0	100.0	0.004	0.136	GSE23984 CD3+T VitaminD analog 10-8M 8days
MGLL	100.0	100.0	0.000	0.034	GSE46268 Monocytes VitaminA 10-8M 18h
MGLL	100.0	100.0	0.001	0.070	GSE46268 Monocytes VitaminA 10-8M 18h
BID	100.0	100.0	0.000	0.006	GSE46268 Monocytes VitaminD 10-8M 18h
BID	100.0	100.0	0.000	0.016	GSE46268 Monocytes VitaminD 10-8M 18h
NETO2	100.0	100.0	0.000	0.016	GSE46268 Monocytes VitaminD 10-8M 18h
BID	100.0	100.0	0.001	0.019	GSE46268 Monocytes VitaminD 10-8M 18h
HMGB3	100.0	100.0	0.001	0.026	GSE46268 Monocytes VitaminD 10-8M 18h

